# Software Requirements Specification for VoiceBridge: An Accessible Speech-to-Control System

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# Revision History

| Date | Version | Notes |
|---|---|---|
| October 2, 2025 | Rawan and Luna | Filled-In Initial Draft of SRS Document |
| October 10, 2025 | Rawan and Luna | Completed SRS Document |
| October 10, 2025 | Mazen and Kelvin | Reviewed and Edited SRS Document |
| November 3, 2025 | Luna | Addressed SRS Glossary TA and Peer Feedback, added customer to glossary for clarity, minor glossary organization improvements |
| December 19, 2025 | Rawan | Addressing peer feedback on caregivers as stakeholders, and addition of C5 and FR6 |
| December 19, 2025 | Rawan | Addressing TA feedback: adding section descriptions, figure/table descriptions, and clarification to varying TA questions, removing duplicate constraint (C1), UML description, and other minor fixes |
| December 24, 2025 | Luna | Addressing TA feedback: adding requirement names, hyperlinks and minor fixes |
| December 25, 2025 | Luna | Addressing Peer feedback: expanding rationale behind choice of 16Hz in FR1 |
| January 10, 2025 | Mazen | Linked SRS to Hazard Analysis Requirements for consistency |

Table 1: Revision History for SRS Document

# 1 Purpose of the Project

## 1.1 User Business

Individuals with speech impairments face significant barriers when interacting with digital devices. VoiceBridge addresses this gap by providing an accurate, inclusive, and accessible speech-to-control system that enables users to communicate with their devices using their speech, regardless of clarity of articulation. Building on familiar technology, such as personal computers and mobile devices, may be one of the most cost-effective and easily adoptable approaches for improving an individual with disability's autonomy and access to the world around them. The rise of automatic speach recognition (ASR) technology and Artificial Intelligence (AI) integrations in the industry provides a novel landscape of opportunities to improve accessibility interfaces. VoiceBridge exploits bleeding-edge technology for a practical and impactful application.

## 1.2 Goals of the Project

| ID | Goal | Description |
|---|---|---|
| G1 | **Accurate Speech Transcription** | Reliably convert impaired or slurred speech into text. |
| G2 | **Command Mapping** | Translate recognized speech into actionable browser commands. |
| G3 | **User Independence** | Enable users with speech impairments to browse autonomously. |
| G4 | **Lightweight & Accessible Design** | Keep the system simple, fast, and cost-effective. |
| G5 | **Cross-Browser Compatibility** | Support major browsers (e.g., Chrome, Edge, Firefox). |
| G6 | **Robust Error Handling** | Detect and recover gracefully from common failures. |
| G7 | **Data Privacy & Security** | Protect user data and ensure secure local processing. |
| G8 | **Customizable Interface** | Allow users to adjust sensitivity, shortcuts, and feedback modes. |
| G9 | **Scalable Architecture** | Design the system for future integration beyond browsers. |

Table 2: Project Goals for VoiceBridge

# 2 Stakeholders

## 2.1 Client

The primary client for the VoiceBridge project is the organization or individual funding or commissioning the system. The client is primarily concerned with achieving the following goals: **G1**, **G5**, and **G9**, ensuring accurate speech transcription, cross-browser compatibility, and scalable architecture.

## 2.2 Customer & Hands-On Users of the Project

The primary customers are also the users of the project, them being individuals with speech impairments, who seek independence and autonomy through technology. Their needs directly relate to: **G1**, **G2**, **G3**, and **G8**.

## 2.3 Other Stakeholders

Secondary stakeholders include experts in linguistics, speech processing, and healthcare domains:

- Speech researchers and linguistics specialists, including the project supervisor, Dr. Christian Brodbeck, who provide insight toward **G1** and **G6**.
- Healthcare professionals and speech therapists who advise on usability and accessibility, contributing to **G3** and **G8**.
- Accessibility advocates and organizations interested in promoting the application, aligned with **G4** and **G5**.
- Software developers who implement and maintain the system, supporting **G9** and **G6**.

Tertiary stakeholders include caregivers and professionals who interact with end users, supporting **G3** and **G7**.

## 2.4 Personas

Potential end users of VoiceBridge include:

- **Amira**, a 45-year-old with Parkinson's disease, uses the system to log into Gmail and send emails.
- **David**, a stroke survivor, uses the system to browse the web and make purchases.

## 2.5 Priorities Assigned to Users

The highest priorities are assigned to end users with speech impairments, as their experience with the system defines its success. Secondary priorities include caregivers and technical experts who support the end users in using and maintaining the system.

## 2.6 User Participation

Individuals matching the target user profiles will be recruited for prototype testing and personalization development. Their participation directly supports refining goals **G1**, **G2**, **G3**, and **G8**.

## 2.7 Maintenance Users and Service Technicians

**Maintenance Users:**

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- **Role:** End-users or caregivers performing basic troubleshooting and initiating support requests.
- **Responsibilities:**
  - Reporting errors or unexpected system behavior (**G6**).

  - Installing application updates (**G9**).
  - Managing user-specific configurations (**G8**).

**Service Technicians:**

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- **Role:** Trained technical staff with deeper access to system logs and back-end services.
- **Responsibilities:**
  - Investigating reported issues (**G6**).
  - Ensuring transcription accuracy (**G1**).
  - Deploying updates and patches (**G9**).
  - Ensuring compatibility with operating systems and accessibility frameworks (**G5**).
  - Performing preventive maintenance, including performance monitoring and optimization (**G1**, **G6**).

### 2.8 Society

Society is recognized as an essential stakeholder in the VoiceBridge project, given the system's potential impact on accessibility, inclusion, and digital independence for individuals with speech impairments. The project directly supports broader social goals of equity and participation by enabling users to interact with technology without barriers, contributing to fair access to digital communication and online services.

**Health and Safety:** VoiceBridge promotes safe technology use by minimizing physical strain associated with traditional input methods, such as typing or clicking, which can be challenging for users with motor limitations. Data privacy and security protocols safeguard user health information and prevent psychological stress caused by data misuse or breaches, aligning with ethical design standards for assistive technology.

**Cultural and Ethical Considerations:** The system is designed to accommodate diverse speech patterns, dialects, and accents, ensuring that cultural and linguistic differences are respected rather than penalized by the model. VoiceBridge avoids stigmatizing language or error messaging, instead emphasizing supportive feedback to maintain user confidence and dignity.

**Social Impact:** By reducing communication barriers, VoiceBridge fosters greater inclusion in workplaces, education, and online communities. The system thus contributes to both individual empowerment and collective social benefit.

## 3 Mandated Constraints

### 3.1 Solution Constraints

**C1 — The product shall run on consumer-grade hardware using a personal computer microphone and support macOS, Windows, and Linux distributions.**
**Rationale:** Requiring only consumer-grade hardware ensures broad accessibility for end users and avoids dependency on specialized equipment.
**Fit Criterion:** The system must successfully be integrated onto browsers and run on laptops or desktops across the supported operating systems, using either built-in or external microphones.

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**C2 — The product shall accept non-deterministic user input in the form of natural language speech.**
**Rationale:** Individuals with speech impairments may produce varied speech patterns that cannot be handled by rigid or deterministic command structures.
**Fit Criterion:** The system must be capable of processing and responding to variable natural language inputs without requiring a fixed set of commands.

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**C3 — The product shall integrate with a browser control application to execute voice-based commands.**
**Rationale:** Browser interaction is a primary accessibility point for most digital services, and browser control is essential for practical use of the system.
**Fit Criterion:** The system must successfully perform browser actions (e.g., opening tabs, navigating to URLs, scrolling) through the integrated control application.

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## 3.2 Implementation Environment of the Current System

**C4 — The product will operate within a consumer computing environment consisting of personal computers equipped with microphones.**
**Rationale:** This setup reflects the most common user hardware configuration, ensuring the solution is accessible without additional devices.
**Fit Criterion:** All core functionalities must operate correctly on standard personal computers with a functioning audio input device.

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## 3.3 Partner or Collaborative Applications

**C5 — The initial integration target is Browser Use, an open-source browser control and automation application.**
**Rationale:** Leveraging existing open-source browser automation tools accelerates development and reduces implementation complexity.
**Fit Criterion:** The system must demonstrate the ability to execute at least three browser actions through the integrated partner application during testing.

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**C6 — Future integrations may include API servers, mobile device agents, and home assistants to extend accessibility and functionality.**
**Rationale:** Ensuring extensibility allows the system to grow and adapt to new platforms or user needs.
**Fit Criterion:** The system architecture must allow seamless integration with additional partner applications without major redesign.

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## 3.4 Off-the-Shelf Software

**C7 — The product will rely on off-the-shelf software components, specifically Browser Use for automation and Project Euphonia for voice recording.**
**Rationale:** Using established open-source components allows focus on core functionality, improves maintainability, and supports experimentation with real-world data.
**Fit Criterion:** These components must be successfully integrated into the development workflow, and collected recordings must be stored and processed for model training without licensing conflicts.

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### 3.5 Anticipated Workplace Environment

**C8 — The product may be used at home, in public spaces, or in clinical settings, each with varying background noise conditions.**
**Rationale:** The product must function effectively across diverse real-world environments to meet accessibility goals.
**Fit Criterion:** The speech recognition system must maintain acceptable accuracy and responsiveness across all three identified environments during testing.

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### 3.6 Schedule Constraints

**C9 — Proof of concept must be completed by November 17, 2025. MVP must be ready within eight months, by May 2026.**
**Rationale:** Deadlines align with capstone project milestones and funding timelines, ensuring timely testing and delivery.
**Fit Criterion:** All proof of concept requirements must be met by the November deadline, and MVP functionality must be fully operational by May 2026.

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### 3.7 Budget Constraints

**C10 — The project must operate within the capstone budget allocated for compute infrastructure and development resources.**
**Rationale:** Budget limitations require prioritizing open-source solutions and efficient resource allocation.
**Fit Criterion:** Total cost of infrastructure, hosting, and third-party services must not exceed the allocated capstone budget.

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### 3.8 Enterprise Constraints

**C11 — The product must comply with all relevant accessibility and privacy regulations, including data usage and user privacy agreements.**
**Rationale:** Compliance protects user rights, upholds institutional standards, and avoids legal or ethical issues.
**Fit Criterion:** All data collection and processing workflows must undergo compliance review, and accessibility features must align with recognized standards (e.g., WCAG).

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# 4 Naming Conventions and Terminology

## Glossary of Terms and Acronyms

### General Project Terms

**Client** Group or individual funding or commissioning the system, typically the organization requesting the software.

**Customer** An individual or organization that purchases, subscribes to, or deploys the VoiceBridge system for use by end users. Customers may not directly participate in defining requirements but represent those who adopt or distribute the final product.

**End User** Someone who directly uses the system to assist with speech or browser tasks.

**Persona** Example character representing a user to demonstrate design needs, such as Amira with Parkinson's or David after a stroke.

**Requirement Phase-In** The process of introducing requirements in planned stages, from proof-of-concept to full deployment.

**Stakeholder** Any individual or group involved in or affected by the project, including users, caregivers, technical experts, and funding organizations.

**VoiceBridge** A software system that enables users with speech impairments to control browsers and devices using their voice, even if their speech is slurred or hard to understand. It is the technology whose requirements are described in this document.

### Technical Terminology

**API (Application Programming Interface)** A set of rules allowing one software program to communicate, exchange services or data with another.

**ASR (Automatic Speech Recognition)** Technology that listens to speech and automatically converts it into written text.

**Browser Agent** Software/code that helps automate browser control tasks.

**CLI (Command Line Interface)** An interaction method using typed text to control a computer, usually for advanced settings.

**JSON (JavaScript Object Notation)** A simple data format both easy for humans and computers to read and write.

**LLM (Large Language Model)** Advanced AI designed for understanding and generating human language.

**Microphone Input** Audio signal collected from a user's microphone for speech processing.

**MVP (Minimum Viable Product)** Basic version of software with enough features for user testing before further development.

**PoC (Proof of Concept)** A small system built to demonstrate that the main idea works in practice.

**STT (Speech To Text)** Another term for ASR, representing the conversion of speech into text.

**Session** The time period when a user is actively interacting with the system, such as from login to logout.

**TTS (Text To Speech)** Technology that transforms written text into spoken words using a computer-generated voice.

**UML (Unified Modeling Language)** Visual diagram style to represent system structure/workflow, mainly for programmers.

**UUID (Universally Unique Identifier)** Special ID assigned to sessions or user interactions for clear data tracking.

### Medical Terminology

**Aphasia** Medical condition causing difficulties in communication, affecting speaking, writing, and understanding language.

**ALS (Amyotrophic Lateral Sclerosis)** Nervous system disease affecting movement, including speech, due to muscle weakness.

**Dysarthria** Speech disorder resulting in slurred/slow speech because of nervous system damage.

**Parkinson's Disease** Progressive neurological disorder affecting movement and speech, often causing tremors and slurred speech.

### Accessibility and Compliance

**Accessibility** Creating software everyone can use, including those with disabilities (e.g., clear text, screen reader support).

**PIPEDA (Personal Information Protection and Electronic Documents Act)** Canadian law regulating personal data collection, use, and protection.

**WCAG (Web Content Accessibility Guidelines)** Guidelines for making web content usable for people with disabilities; "Level AA" meets many key needs.

### System Features and Components

**Audit Logging** Recording system actions/events to track problems or security issues.

**Command Mapping** System's ability to recognize and map transcribed text to actions (e.g., "open Gmail").

**Data Encryption** Protecting information by converting it into unreadable code except for authorized parties.

**Error Handling** Detecting, reporting, and recovering from mistakes or unexpected situations.

**Latency** Time between user speech input and system response.

**Personalization** Adjusting system behavior to match each user's preferences, speech pattern, or needs.

**Robustness/Fault-Tolerance** Ability to keep working even with errors, noise, or network issues.

**Scalability** Ability for a system to support more users or new features over time.

**Transcription Accuracy** How correctly speech is transformed into text; higher accuracy means fewer mistakes.

### Browser and Platform Terms

**Browser Control Application** Tool for users to control browsers with voice commands.

**Cross-Browser Compatibility** Ability for software to work across different browsers (Chrome, Edge, Firefox, etc.).

**Device Command** Instruction sent to a computer to perform a task, such as opening applications or typing.

**Off-the-Shelf Software** Ready-made software/components integrated into the project instead of building from scratch.

### Miscellaneous Terms

**Data Dictionary** Documentation listing information types used by the system, with explanations.

**Feedback Message** Confirmation or notification shown to users, indicating actions or errors.

**Intent** The underlying meaning or goal of what the user says, interpreted by the system.

**Training Data** Audio recordings and transcripts used to improve speech recognition accuracy.

**User Profile** Data associated with each user, like preferences and speech patterns, for personalized command recognition.

### Acronym Reference Table

Table 3 provides a consolidated list of all acronyms used throughout this document.

| Acronym | Full Term | Simple Description |
|---|---|---|
| ALS | Amyotrophic Lateral Sclerosis | Disease causing muscle movement problems. |
| API | Application Programming Interface | Rules for different software to work together. |
| ASR | Automatic Speech Recognition | Converts spoken words into text automatically. |
| BUC | Business Use Case | Description of a scenario representing how a business or organization will use the system. |
| CER | Character Error Rate | Metric for evaluating speech recognition accuracy at the character level, useful for fine-grained analysis. |
| CLI | Command Line Interface | Computer control by typing commands. |
| FAQ | Frequently Asked Questions | Common questions and answers list. |
| JSON | JavaScript Object Notation | Easy data format for computers and humans. |
| LLM | Large Language Model | Advanced AI for understanding language. |
| MVP | Minimum Viable Product | First usable version with basic features. |
| PIPEDA | Personal Info Protection Document Act | Canadian privacy protection law. |
| PoC | Proof of Concept | A small-scale early prototype. |
| STT | Speech To Text | Converts speech into written text. |
| TTS | Text To Speech | Converts text to computer-generated spoken words. |
| UML | Unified Modeling Language | Visual diagrams for system planning. |
| UUID | Universally Unique Identifier | Special ID for sessions or users. |
| WCAG | Web Content Accessibility Guidelines | Rules for making web sites accessible. |
| WER | Word Error Rate | Metric for evaluating the accuracy of speech recognition by comparing transcribed words to the reference text. |

Table 3: Acronym Quick Reference

# 5 Relevant Facts And Assumptions

## 5.1 Relevant Facts

| Fact ID | Fact | Explanation / Relevance |
|---|---|---|
| F1 | **Users experience speech impairments of varying severity.** | VoiceBridge must handle varying speech clarity, from slurred to partially formed words (related to (**G1**), (**G2**)). |
| F2 | **Users want to communicate and navigate independently.** | Motivates autonomy in using the system without reliance on caretakers (supports (**G3**)). |
| F3 | **Users may have limited mobility.** | Hands-free operation improves accessibility and inclusion (supports (**G3**), (**G4**)). |
| F4 | **Users value simplicity and low cognitive load.** | Commands should remain intuitive and easy to use (supports (**G4**)). |
| F5 | **Users may have emotional sensitivity around speech difficulty.** | Interface should be respectful and encouraging without repeating goals (complements (**G6**)). |
| F6 | **Users expect privacy and dignity.** | Data handling should preserve privacy and transparency (supports (**G7**)). |
| F7 | **Users may use different languages or accents.** | System must accommodate linguistic diversity without bias (relates to (**G1**), (**G8**)). |
| F8 | **Users may use assistive tools concurrently.** | VoiceBridge should integrate seamlessly with other accessibility tools (supports (**G4**), (**G8**)). |
| F9 | **Users will vary in technical comfort.** | Onboarding should be minimal and low-friction (supports (**G4**), (**G8**)). |
| F10 | **Users appreciate visual feedback and control.** | Feedback builds trust and reduces frustration (complements (**G1**), (**G8**)). |

Table 4: User-Centered Facts for VoiceBridge

## 5.2 Business Rules

| ID | Business Rule | Rationale |
|---|---|---|
| BR1 | Users must be able to cancel or stop a command at any time. | Empowers users and prevents frustration. (G3, G4) |
| BR2 | Transcribed text must be displayed for user verification before executing critical commands. | Ensures accuracy, avoids misinterpretation, and maintains user confidence, inline with (G1) |
| BR3 | Browser commands must not execute without user consent for actions with potential data impact (e.g., sending messages, closing tabs). | Protects user privacy and prevents accidental operations. (G2, G7) |
| BR4 | The system should provide immediate visual feedback within 5 seconds of speech input. | Builds trust, transparency, and usability for impaired users. (G6, G1) |
| BR5 | System must handle moderate background noise without significant degradation of performance. | Maintains reliability in real-world environments.(G1, G6) |
| BR6 | Users should be able to configure simple command mappings for personalized tasks. | Supports individual preferences and improves user autonomy. (G2, G3, G8) |

Table 5: Business Rules for VoiceBridge

### 5.3 Assumptions

| ID | Assumption | Implication for Design |
|---|---|---|
| A1 | **Users have access to a working microphone and modern browser.** | The system assumes functional input hardware and browser APIs for speech capture (see G1, G5). |
| A2 | **Users will tolerate minor transcription errors if quickly correctable.** | Fast feedback and correction options are more important than perfect accuracy (see G1, G6). |
| A3 | **Users are willing to train or calibrate the model briefly.** | A short setup phase (e.g., sample phrases) can improve recognition quality (see G1, G8). |
| A4 | **Users prefer transparent, explainable behavior.** | VoiceBridge should indicate what command is being executed to prevent confusion or mistrust (see G8). |
| A5 | **Users may be in noisy or uncontrolled environments.** | Noise-robust models and confirmation prompts are required to maintain reliability (see G1, G6). |
| A6 | **Users want emotional ease of use.** | Tone and interface language must feel supportive — e.g., "Let's try again" instead of "Error" (see G6). |
| A7 | **Users will likely use the tool for daily web tasks.** | The feature set should prioritize essential browser actions (navigation, scrolling, typing, tab control) (see G2, G3, G8). |

Table 6: Design Assumptions for VoiceBridge

## 6 The Scope of the Work

### 6.1 The Current Situation

Individuals with speech impairments currently rely on standard speech-to-text or manual input to use browsers and devices. Existing systems struggle with slurred or atypical speech, often requiring caregiver assistance.

Typical workarounds include typing commands, using alternative inputs, or correcting errors, leading to frustration and limited autonomy.

Current workflow: User speaks → Standard recognition → Frequent errors → Manual/caregiver correction → Action executed

VoiceBridge aims to replace this with accurate transcription, command mapping, and immediate feedback, enabling independent browser control.

## 6.2 The Context of the Work

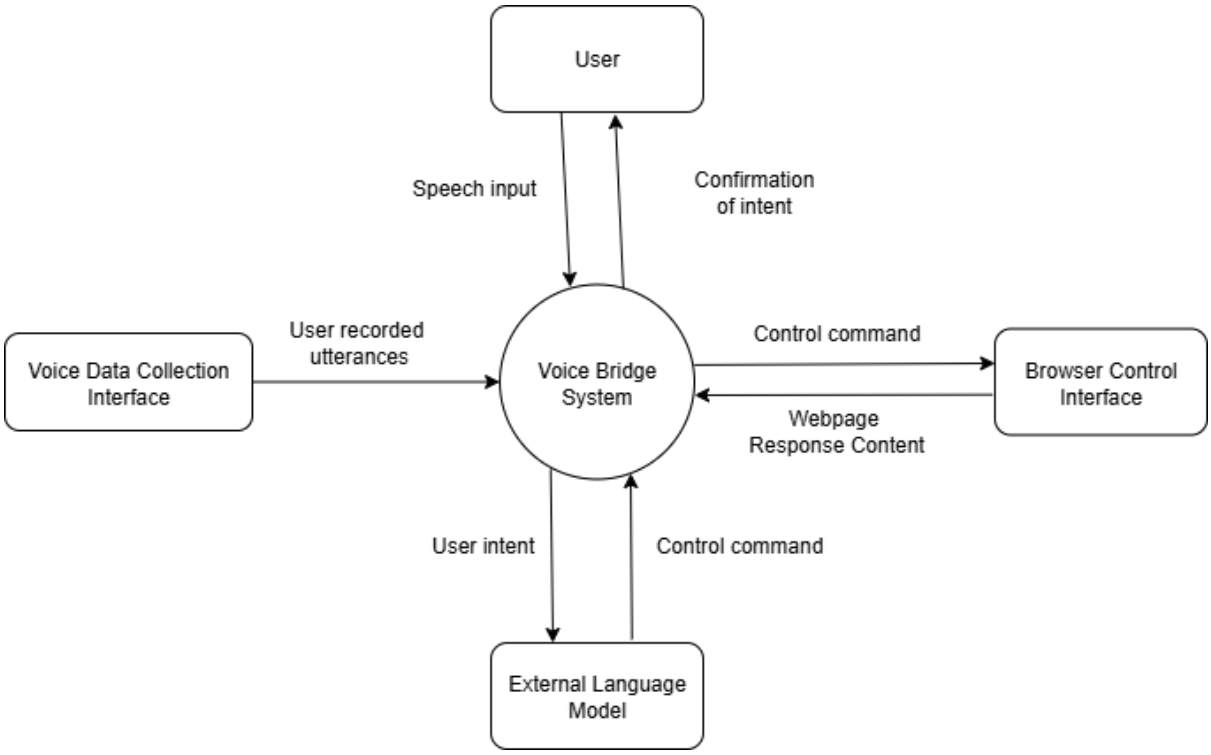


Figure 1: Context Diagram of VoiceBridge

## 6.3 Work Partitioning

| Event Name | Input | Output | Brief BUC Summary | Relevant Data Classes |
|---|---|---|---|---|
| **Input Capture** | User speech via microphone | Audio stream to system | Capture speech for processing | Raw audio, timestamp |
| **Speech-to-Text Modelling** | Audio stream | Transcribed text | Convert impaired speech to text | Audio data, transcription |
| **Intent Confirmation** | Transcribed text | User confirmation | Verify intended command | Text data, confidence score |
| **Command Mapping** | Confirmed text | Actionable command | Map text to browser action | Command definitions, user preferences |
| **Browser Interaction Layer** | Actionable command | Executed browser action | Perform command in browser | Command data, page context, execution status |

Table 7: VoiceBridge Events, Inputs, Outputs, and Data Classes

### 6.4 Specifying a Business Use Case (BUC)

| BUC ID | Name | Description |
|---|---|---|
| BUC-1 | Capture Speech Input | The user initiates a speech session, and the system captures audio from the microphone for ASR processing. |
| BUC-2 | Transcribe Speech to Text | The system converts captured audio into text through feature extraction and ASR model inference. |
| BUC-3 | Interpret Intent | The system parses recognized text to determine the user's intent and prepares a structured intent object. |
| BUC-4 | Execute Browser Command | The structured intent is mapped to a browser action, executed, and confirmed visually or audibly to the user. |

Table 8: Business Use Cases (BUCs) for VoiceBridge

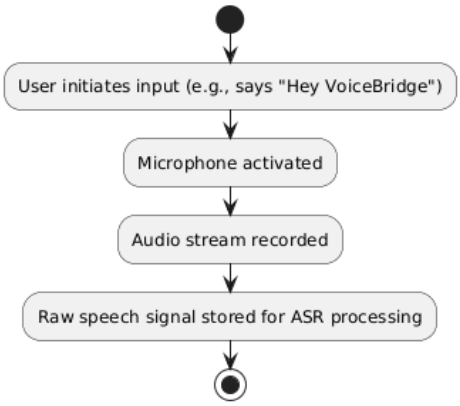


Figure 2: BUC-1: Capture Speech Input

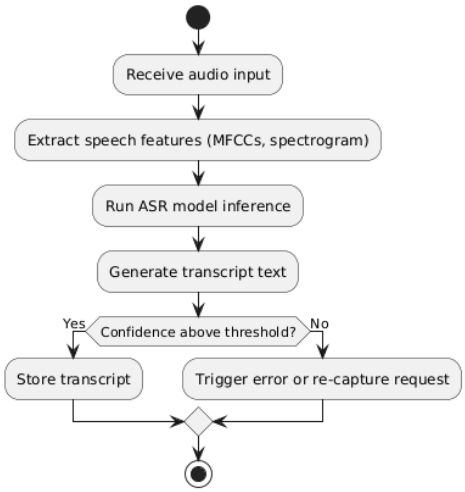


Figure 3: BUC-2: Transcribe Speech to Text

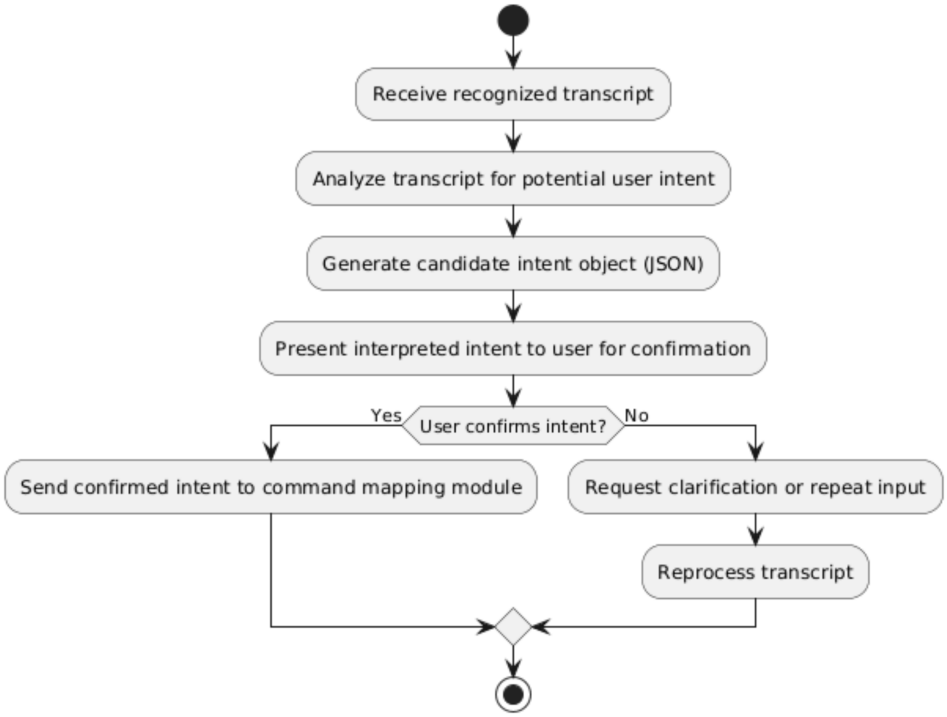


Figure 4: BUC-3: Interpret Intent

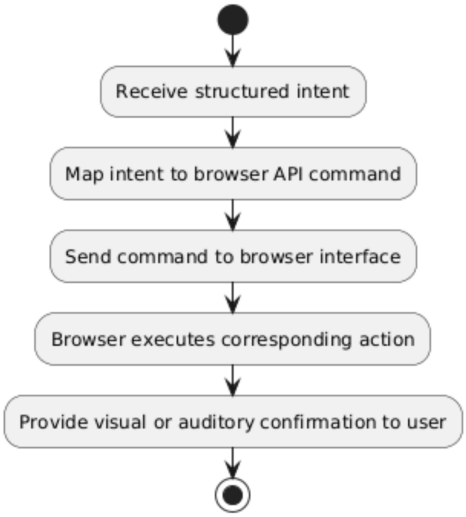


Figure 5: BUC-4: Execute Browser Command

# 7 Business Data Model and Data Dictionary

## 7.1 Business Data Model

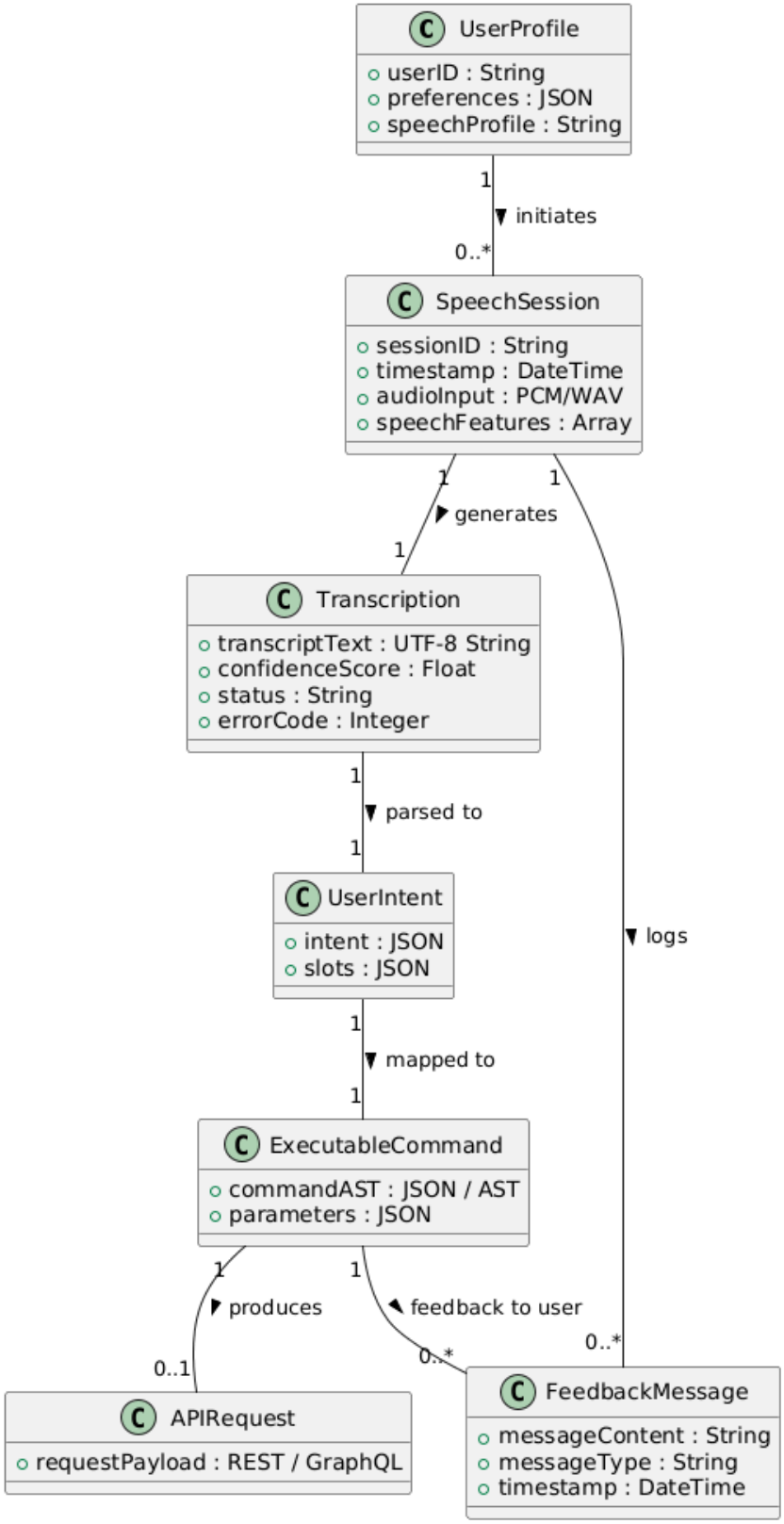


Figure 6: VoiceBridge UML Diagram.

## 7.2 Data Dictionary

| Field | Description | Type | Source | Usage |
|---|---|---|---|---|
| **Audio_Input** | Raw speech signal captured from microphone | PCM, WAV, 16 kHz, 16-bit | User microphone | Primary input for ASR processing |
| **Speech_Features** | Extracted acoustic features (e.g., MFCCs, spectrogram) | N-dimensional array | Feature extractor module | Internal representation used by ASR engine |
| **Transcript_Text** | Recognized text output from speech input | UTF-8 encoded free text | ASR engine | Basis for interpreting user intent |
| **Error_Code** | Status or error indicator for system response | Integer | ASR engine / Controller | 0 = Success, ¿0 = specific error type |
| **User_Profile** | Speaker-specific data such as ID, preferences, and speech patterns | JSON object | User database | Enables personalized recognition and response |
| **Timestamp** | Time marker for recognition or command event | DateTime | System clock | Used for logging, debugging, and tracking |
| **User_Intent** | Parsed intent derived from natural language input | JSON (intent and slots) | Interpreter module | Structured meaning representation (e.g., {"intent": "open_file", "file": "report.pdf"}) |
| **Exec_Command** | Structured command before browser execution | JSON object or Abstract Syntax Tree | Interpreter module | Translates user intent into executable browser actions |
| **API_Request** | Outgoing request to external or browser API | REST / GraphQL formatted message | Interpreter module → API translator | Executes command or triggers action in external system |
| **Feedback_Message** | User-facing feedback or error output | String / Audio / JSON | VoiceBridge UI | Confirms action success or requests clarification |
| **Session_ID** | Unique identifier for a single user interaction session | UUID / String | System controller | Links data objects across a single speech-to-command event |

Table 9: VoiceBridge Data Dictionary

# 8 The Scope of the Product

## 8.1 Product Boundary

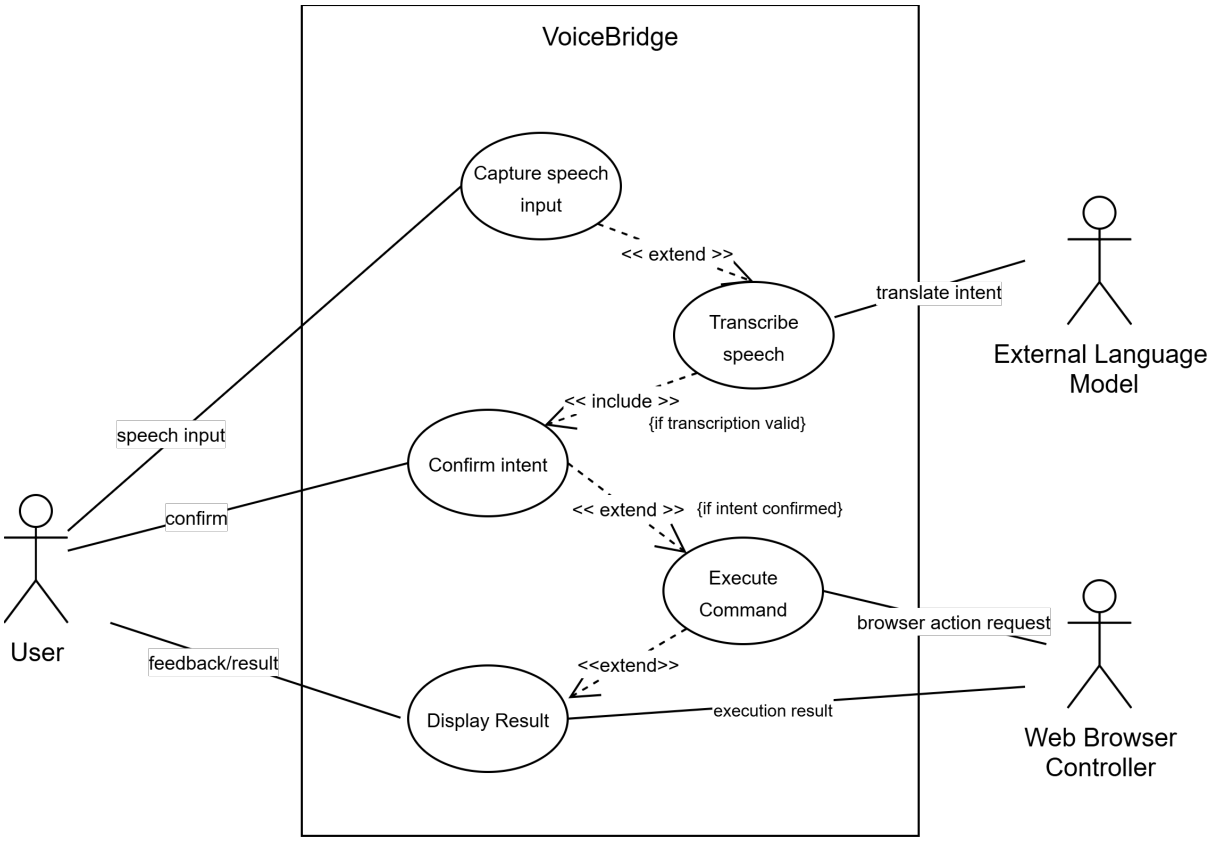


Figure 7: VoiceBridge Use Case Diagram. Each ellipse represents a Product Use Case (PUC) corresponding to a major system function.

## 8.2 Product Use Case Table

Table 10: Product Use Case (PUC) Table

| PUC ID | Event Trigger | Input | Output | Description |
|---|---|---|---|---|
| PUC-1 | User speaks into microphone | Audio signal | Captured speech data | System listens for user speech input. |
| PUC-2 | Speech captured | Speech data | Text transcription | Converts impaired speech to text using trained model. |
| PUC-3 | Transcription complete | Transcribed text | Confirmation prompt / feedback | Seeks user confirmation to ensure correct interpretation. |
| PUC-4 | Intent confirmed | Confirmed command text | Browser command execution | Maps intent to actionable browser function. |
| PUC-5 | Command executed or error | System response | Success or error feedback to user | Provides feedback or prompts retry on failure. |

### 8.2.1 Individual Product Use Cases (PUC's)

This section defines the detailed Product Use Cases (PUCs) for the VoiceBridge system. Each PUC describes the interaction between the user and the system, including purpose, actors, triggers, inputs/outputs, and main scenarios.

**PUC-1: Capture User Speech**

**Primary Actor:** User
**Trigger:** User begins speaking into the microphone.
**Precondition:** The microphone is connected and permissions are granted.
**Input:** Audio signal (live speech).
**Output:** Captured audio data stream.
**Postcondition:** Audio data is made available for processing by the ASR (Automatic Speech Recognition) module.

**Main Scenario:**

1. User activates the VoiceBridge interface (e.g., presses a "Listen" button).
2. The system listens through the microphone for input.
3. The system captures the speech signal and stores it temporarily in memory.
4. Captured data is sent to the speech-to-text module for processing.

**Alternative Flow:**

- If the microphone is unavailable or access is denied, the system displays an error prompt.
- User can retry after adjusting permissions or hardware connection.

**PUC-2: Convert Speech to Text**

**Primary Actor:** System (Speech Recognition Module)
**Trigger:** Audio capture event completed.
**Input:** Captured speech data.
**Output:** Transcribed text.
**Postcondition:** Transcription results are ready for user confirmation.

**Main Scenario:**

1. System processes the captured audio stream using a trained ASR model.
2. Acoustic and linguistic features are extracted.
3. Speech is converted to text and stored temporarily.
4. Transcribed text is passed to the confirmation display.

**Alternative Flow:**

- If transcription confidence is below threshold, the system requests a repeat.

**PUC-3: Confirm Transcription**

**Primary Actor:** User
**Trigger:** System displays transcribed text.
**Input:** Transcribed text.
**Output:** User confirmation or correction.

**Main Scenario:**

1. The transcribed text is displayed to the user for verification.
2. User confirms that the text is correct or requests reprocessing.
3. System records the confirmation and proceeds to intent mapping.

**Alternative Flow:**

- If user rejects the transcription, the system returns to PUC-1 for re-input.

**PUC-4: Map Intent to Browser Command**

**Primary Actor:** System (Command Mapping Module)
**Trigger:** User confirmation received.
**Input:** Confirmed command text.
**Output:** Actionable browser command.

**Main Scenario:**

1. System parses confirmed text for intent (e.g., "open YouTube").
2. System searches for matching browser or OS command.
3. Mapped command is passed to the execution layer.

**Alternative Flow:**

- If no matching command is found, the system provides suggestions.

**PUC-5: Execute Command and Provide Feedback**

**Primary Actor:** System
**Trigger:** Actionable command received.
**Input:** Command representation (API or accessibility call).
**Output:** Visible browser or OS action, and feedback message.

**Main Scenario:**

1. System executes the command using the browser's API or accessibility layer.
2. The target application or tab performs the intended action (e.g., opens a webpage).
3. System provides visual and/or auditory feedback to confirm success.

**Alternative Flow:**

- If the command execution fails, an error message or retry option is displayed.

# 9 Undesired Event Handling

| Event ID | Description | Possible Cause | System Response / Handling Strategy | Severity | User Notification |
|---|---|---|---|---|---|
| UE-1 | No Speech Detected | User is silent or microphone muted. | System times out after configurable interval (e.g., 5 s) and displays "No input detected. Please speak again." | Low | Visual and optional audio cue. |
| UE-2 | Unintelligible Speech / Low Confidence Transcription | Background noise, unclear articulation, or poor model match. | VoiceBridge requests repetition and highlights uncertain words. Logs instance for model improvement. | Medium | On-screen prompt: "Sorry, I didn't catch that." |
| UE-3 | Misinterpreted Command | Incorrect speech-to-text mapping or ambiguous command phrase. | Requires confirmation before executing potentially disruptive browser actions (e.g., closing tab, submitting form). | High | Confirmation dialog: "Did you mean 'Close tab'?" |
| UE-4 | Microphone Access Denied | User or OS blocks microphone permission. | System notifies user and provides step-by-step instructions to enable microphone in browser/OS settings. | High | Visual warning banner. |
| UE-5 | Browser Extension / API Failure | Browser crashes, extension disabled, or interface unresponsive. | Detect failure via heartbeat signal. Attempt reconnection; if unsuccessful, disable commands and show recovery notice. | High | "Browser connection lost — attempting to reconnect." |
| UE-6 | Network Connectivity Loss | Internet connection interrupted. | Offline mode activated: local commands (scroll, zoom, etc.) remain functional; external navigation disabled. | Medium | "Network unavailable — limited mode active." |
| UE-7 | System Crash / Model Process Failure | Unexpected error during ML inference or memory overflow. | System logs event, restarts inference module, and recovers previous session state when possible. | Critical | "VoiceBridge restarted after an error." |
| UE-8 | Data Privacy Breach Attempt | Unauthorized access to stored speech logs or transcripts. | Encrypts all local logs; triggers security alert, terminates unauthorized session, and notifies user. | Critical | "Security alert — your data is protected and the session has been closed." |
| UE-9 | Latency Exceeds Threshold | Inference or browser response exceeds acceptable delay (e.g., $> 2$ s). | Display progress indicator; optionally degrade to simplified model or local processing fallback. | Low | Spinner with "Processing..." message. |
| UE-10 | User Cancels Ongoing Command | User says "cancel" or "stop." | Immediately halts current command execution and clears pending queue. | Low | Confirmation sound or text "Cancelled." |

Table 12: VoiceBridge User Event Handling Scenarios

# 10 Functional Requirements

## 10.1 Functional Requirements

## 10.2 FR1: Accept Speech Audio via Microphone

**Description:** The system must capture live speech input from the user through a standard built-in or external microphone.
**Rationale:** Speech audio contains frequency components up to approximately 8 kHz. To capture this audio accurately without distortion, the sampling rate must be at least 2× the maximum frequency, resulting in a minimum required rate of 16 kHz. This ensures reliable speech transcription and is supported by standard built-in and external microphone hardware.
**Fit Criterion:** The system reliably detects and records audio from default OS microphone devices across Windows, macOS, iOS, and Android, with a minimum 16 kHz sampling rate.

## 10.3 FR2: Convert Impaired Speech to Text with greater than or equal to 80% Accuracy (MVP)

**Description:** The system must process the captured audio and output a textual representation of the spoken utterance.
**Rationale:** Accurate transcription is the core functionality that enables communication and command execution. Without acceptable accuracy, the product fails its purpose.
**Fit Criterion:** In evaluation on a test dataset of impaired speech, transcription accuracy must reach at least 80% word error rate (WER) reduction compared to baseline models, and achieve greater than or equal to 80% accuracy for common commands.

## 10.4 FR3: Display Transcription for Verification

**Description:** The transcribed text must be displayed in real time on the user's device interface.
**Rationale:** Transparent feedback allows the user to verify correctness, catch errors, and build trust in the system.
**Fit Criterion:** Every spoken input is displayed within 2 seconds as text on the UI, with at least 95% consistency across trials.

## 10.5 FR4: Map Text to Arbitrary Device Commands

**Description:** The system must recognize when transcribed text corresponds to a predefined device action (e.g., "open amazon.ca," "draft a new email") and translate it into the appropriate command representation (API, CLI, or accessibility call).
**Rationale:** Mapping allows the system to extend beyond communication into real device control, empowering user independence.
**Fit Criterion:** For a test set of 50 predefined commands, the system maps user input to the correct command representation in at least 90% of cases.

## 10.6 FR5: Execute Commands on the Host Device

**Description:** The system must execute the mapped commands through the host device's accessibility framework or APIs, resulting in visible user action (e.g., app launch, scrolling, text entry).
**Rationale:** Without execution, the system remains a transcription tool only. Execution closes

the loop between speech input and device interaction.
**Fit Criterion:** For each correctly recognized command, the intended system action occurs on the device within 2 seconds, with greater than or equal to 95% reliability across test scenarios.

### 10.7 FR6: Handle Security Software Interference

**Description:** The system must detect when browser, operating system, or antivirus security controls block or restrict automated actions and provide clear feedback to the user, including the affected command and available fallback options.
**Rationale:** Security software may prevent automated execution, which could otherwise result in silent failures and unsafe user assumptions about task completion. User awareness preserves safety and usability.
**Fit Criterion:** In test scenarios where automation is blocked, the system correctly detects the restriction and notifies the user in greater than or equal to 95% of cases within 2 seconds of the failed action.

## 11 Look and Feel Requirements

### 11.1 Appearance Requirements

**AR1: Minimal and Non-Intrusive Interface**

The interface shall have a clean and minimal design to reduce cognitive load. Key elements should be visually distinct, with consistent spacing, and color usage to support quick recognition of actions.

Since it's a browser integration, the user interface should minimally interfere with the visibility of the content on the page. The interface should only capture the user's attention as functionally needed (i.e., listening to user prompts, confirming user intent), but should otherwise blend in with the browser interface.

### 11.2 Style Requirements

**SR1: Professional Visual Style**

The system shall maintain a professional and neutral visual style suitable for general workplace use. Colors, icons, and fonts should prioritize clarity over branding at this stage. Future iterations may incorporate custom styling or theming.

## 12 Usability and Humanity Requirements

### 12.1 Ease of Use Requirements

**EU1: Low Interaction Effort**

The system should minimize user effort by providing a simple, intuitive interface. Key actions should be accessible within 3-4 interactions, with clear feedback after each action.

### 12.2 Personalization and Internationalization Requirements

**PI1: Personalization and Language Support**

The system should support basic personalization (e.g., remembering user preferences) and allow easy adaptation for different languages or regions at a later stage. For the PoC, English support is sufficient.

### 12.3 Learning Requirements

**LR1: Rapid Learnability**

The system should be learnable within 10 minutes without prior training or documentation. Users should be able to complete core tasks on their first attempt through the interfaces navigation tutorial upon first time launch. Additional documentation should be supplemental but not necessary.

### 12.4 Understandability and Politeness Requirements

**UP1: Clear System Communication**

The system should use clear, direct, and neutral language in responses. Error messages or clarifications should remain polite and informative.

### 12.5 Accessibility Requirements

**AC1: Assistive Tool Compatibility**

The interface should be navigable using standard assistive tools and offer clear text contrast and legible font sizes. Full accessibility compliance is not required at the PoC stage but should be feasible for future iterations.

## 13 Performance Requirements

### 13.1 Speed and Latency Requirements

**PR1: Bounded Speech-to-Action Latency**

Under normal operating conditions, latency requirements can be broken down into:
Speech interpretation: 5 s after end of utterance
Command generation & execution: 10-15 seconds after end of speech interpretation

### 13.2 Safety-Critical Requirements

**PR2: Prevention of Unsafe Actions**

The system shall enforce guardrails to prevent unsafe or unintended actions, requiring validation and user confirmation for potentially disruptive operations, and provide warnings or fail-safes for errors.

### 13.3 Precision or Accuracy Requirements

**PR3: Minimum Recognition Accuracy**

ASR accuracy shall be at least 70% in stationary noise conditions. The system shall achieve at least 80% command recognition precision under stationary noise conditions for the PoC.

### 13.4 Robustness or Fault-Tolerance Requirements

**PR4: Degradation Under Failure**

The system shall remain stable under fluctuating network conditions and noisy input. Fallback mechanisms (e.g., retry logic, error messaging) shall ensure graceful issue handling.

### 13.5 Capacity Requirements

**PR5: Concurrent User Support**

The system shall support at least 20 concurrent users without service degradation if the product is commercialized.

### 13.6 Scalability or Extensibility Requirements

**PR6: Scalable Architecture**

The system architecture shall allow horizontal scaling to handle increased traffic and modular extensions fo new interaction capabilities.

### 13.7 Longevity Requirements

**PR7: Long-Term Operability**

The system shall be designed to operate reliably over a minimum of 5 years, with maintainable and updatable components to support long-term product evolution.

## 14 Operational and Environmental Requirements

### 14.1 Expected Physical Environment

**OE1: Operation in Typical Work Environments**

The product shall be operable in a variety of typical office, home, or institutional environments where users perform their daily tasks. The system shall be robust to stationary background noise such as air conditioning, computer fans, and ambient hum. It is not required to reliably handle non-stationary noise, including multiple people speaking or sudden loud interruptions. No modifications to the host operating system, browser, or network configuration shall be required.

### 14.2 Wider Environment Requirements

**OE2: Web-Based Deployment Environment**

The primary interface shall be web-based, accessible via standard web browsers, to maximize user accessibility and support flexible use cases.

### 14.3 Requirements for Interfacing with Adjacent Systems

**OE3: Integration with External Systems**

1. The system shall integrate with existing browser-based platforms and may interface with external language interpreter modules.
2. Open-source components (e.g., browser interaction agents and libraries) may be incorporated, ensuring compatibility and maintainability.

### 14.4 Productization Requirements

**OE4: Multi-User Deployment and Distribution**

1. The product shall be deployable for multiple users within an organization, supporting secure user accounts and personalized ASR profiles.
2. The design shall allow packaging and distribution without requiring technical setup by end users. The product shall have straightforward installation or access via pre-configured web access and automatic model initialization.
3. Productization shall include logging suitable for monitoring performance and usage in a hosted environment.
4. The system shall include mechanisms for updates to features with minimal disruption to users.

### 14.5 Release Requirements

**OE5: Defined Release Cycle**

The product shall follow a defined release cycle, providing minor updates quarterly and major updates semi-annually. Each release must maintain backward compatibility with user data, personalization settings, and existing features. Release planning will account for maintenance effort, compute resources, and compliance obligations.

## 15 Maintainability and Support Requirements

### 15.1 Maintenance Requirements

**MS1: Maintainable Codebase**

Code must be modular, documented, and testable to support scalability, debugging, and future updates. Maintenance must be possible by developers who were not the original authors.

### 15.2 Supportability Requirements

**MS2: User Support Resources**

The product shall provide an accessible Help Page and a Frequently Asked Questions (FAQ) section with clear instructions. It shall be displayed with high-contrast visuals, simple language, and auditory and visual aids.

### 15.3 Adaptability Requirements

**MS3: Platform Compatibility**

The system must run on common workplace platforms via a web browser, including Windows 10 or later, macOS 12 Monterey or later, and Linux distributions such as Ubuntu 20.04 LTS or later, supporting modern web browsers.

## 16 Security Requirements

The following security/safety requirements are derived from the project Hazard Analysis (see Hazard Analysis Section 6.) Each item is traceable to hazards in the FMEA table and is included here to ensure consistency. We group VoiceBridge's security requirements into five categories—**Access (ACR)**, **Integrity (IR)**, **Privacy (PRR)**, **Audit (ADR)**, and **Immunity (IMR)**.

### 16.1 Access Requirements

**ACR1:** Important actions (like sending data or external messages) must use a clear, two-step confirmation to prevent mistakes. *(links to IR5/IMR7 — unclear confirmation)*

**ACR2:** Access to stored audio or transcripts is restricted by role-based permissions so only authorized users can view or edit sensitive data. *(links to Storage — Private data leaked)*

**ACR3:** Multi-factor authentication (MFA) is required for all administrator or privileged logins to prevent unauthorized entry. *(links to Auth/Access — Unauthorized entry)*

### 16.2 Integrity Requirements

**IR1:** VoiceBridge auto-detects microphone devices and alerts users if permissions are missing, offering a quick way to select a new device. *(links to Input Devices — Wrong mic chosen)*

**IR2:** Noise suppression must run before recognition; when confidence is low, users must confirm results. *(links to Input Devices — Too much background noise)*

**IR3:** The system tracks model accuracy and prompts retraining or calibration when accuracy drops below a set level. *(links to ML Model — Doesn't adapt to accent or changes)*

**IR4:** If the connection is lost, the app shows an offline message, pauses risky actions, and retries once the network returns. *(links to Communication — Weak or lost network)*

**IR5:** Confirmation screens and the overall user interface must be clear, accessible, and perceivable by text, audio, and assistive tools. *(links to Front-end Interface — Confirmation not clear)*

### 16.3 Privacy Requirements

**PRR1:** All stored data is encrypted with restricted access to protect privacy. *(links to Storage — Private data leaked)*

**PRR2:** Logs never store personal or raw speech data; only anonymized and redacted information is kept. *(supports IR2, PRR2)*

**PRR3:** Sessions expire automatically and are revoked if suspicious activity is detected. *(extends ACR3 — unauthorized access)*

**PRR4:** Users can request deletion of stored data; the system removes it after a defined time. *(future privacy maintenance)*

### 16.4 Audit Requirements

**ADR1:** The system records recognition accuracy over time and flags when results decline. *(links to ML Model — Accuracy drops)*

**ADR2:** Keep structured, redacted logs for reviews and debugging. *(links to Storage — Sensitive logging)*

**ADR3:** Create a Software Bill of Materials (SBOM) and block unsafe dependencies. *(links to library risk, supports IMR6)*

**ADR4:** Provide regular privacy reports showing data deletion and retention. *(future extension)*

### 16.5 Immunity Requirements

**IMR1:** The app recovers smoothly from device or permission changes without restarting. *(links to Input Devices — Wrong mic chosen)*

**IMR2:** Maintain good recognition accuracy even with background noise at normal levels. *(links to Input Devices — Too much noise)*

**IMR3:** The app retries failed network actions and cancels incomplete ones to handle weak connections gracefully. *(links to Communication — Weak or lost network)*

**IMR4:** Detect and drop repeated or expired requests to avoid replay attacks. *(future enhancement)*

**IMR5:** Ensure all confirmations can be seen or heard, including through screen readers. *(links to Front-end Interface — not perceivable)*

# 17 Cultural Requirements

## 17.1 Cultural Requirements

**CR1: Culturally Neutral Interaction**

The system shall maintain a culturally neutral and respectful tone when prompting users, avoiding slang, bias, and discriminatory language. It must include ethical guardrails to prevent the generation of harmful content or execution of potentially dangerous commands.

It shall support inclusive and accessible design to serve users across diverse cultural backgrounds.

# 18 Compliance Requirements

## 18.1 Legal Requirements

**CO1: Legal Compliance**

The project shall comply with the Personal Information Protection and Electronic Documents Act (PIPEDA) regarding the collection, storage, and handling of personal information.

## 18.2 Standards Compliance Requirements

**CO2: Accessibility Standards Compliance**

The application must comply with the Web Content Accessibility Guidelines (WCAG) 2.0, Level AA guidelines to ensure usability by individuals with disabilities.

# 19 Requirements Likely and Unlikely to Change

## 19.1 Likely to Change

- **12.1 Speed & Latency:** The system shall process speech input and return text in near real-time.
  *Rationale:* Performance targets may improve as speech recognition and browser processing capabilities evolve.

- **14.3 Adaptability:** The system shall allow updates for new languages, user preferences, or accessibility features.
  *Rationale:* Future user needs or new assistive technologies may require system modifications.

- **11.2 Personalization & Internationalization:** The system shall support preferred voice, accent handling, and language selection.
  *Rationale:* User preferences and supported languages are likely to expand over time.

- **12.6 Scalability & Extensibility:** The system shall support an increasing number of concurrent users and new modules.
  *Rationale:* User growth or additional features may require changes in infrastructure.

- **13.5 Release Requirements:** Updates shall be deployable on web browsers without downtime.
  *Rationale:* Deployment strategies may change as development tools or environments evolve.

- **26.2 Personalized ASR Fine-Tuning:** The system shall allow individualized model tuning to improve transcription accuracy.
  *Rationale:* Model improvements and user feedback may necessitate adjustments.

- **17.2 Standards Compliance Requirements:** System shall follow relevant industry standards.
  *Rationale:* Standard adherence ensures safety, reliability, and credibility. As the product evolves, new features may require compliance with additional standards.

### 19.2 Unlikely to Change

- **9.1 Functional Requirements:** The system shall accurately transcribe speech to text for users with speech impairments.
  *Rationale:* Core transcription functionality is foundational and will not change.
- **10.1 Appearance:** Interface shall be clean, minimal, and visually organized to reduce cognitive load.
  *Rationale:* Minimal and clear design is essential for accessibility and usability.
- **10.2 Style:** Interface shall use consistent fonts, spacing, and colors to support clarity.
  *Rationale:* Visual consistency is fundamental for usability.
- **11.1 Ease of Use:** Users shall complete tasks with minimal steps or cognitive effort.
  *Rationale:* Streamlined interaction is a core accessibility requirement.
- **15.3 Privacy Requirements:** User data shall be encrypted in the database and logs.
  *Rationale:* Privacy protection is legally required and critical for trust, this requirement will not change.
- **17.1 Legal Requirements:** System shall comply with relevant regulations (e.g., PIPEDA).
  *Rationale:* Compliance is mandatory for this project, must be complied with at all times.

## 20 Open Issues

The primary open issue is maintaining high ASR accuracy across the diverse and severe spectrum of dysarthric speech. This requires extensive training and validation data, as well as careful model tuning.

## 21 Off-the-Shelf Solutions

### 21.1 Ready-Made Products

The project will evaluate existing, specialized speech recognition models and applications (e.g., Whisper model and Project Euphonia) as performance baselines.

### 21.2 Reusable Components

Potential reused components could include existing Text-to-Speech (TTS) modules for feedback and LLMs to map user commands into structured actions.

### 21.3 Products That Can Be Copied

Open-source browser automation agents (e.g., The AI browser agent) may be integrated for command execution via voice input.

# 22 New Problems

## 22.1 Effects on the Current Environment

The product shall operate without modifying the user's OS, browser, or network configuration. This separation prevents unintended impact from incorrect commands.

## 22.2 Effects on the Installed Systems

The product shall not bypass firewalls, alter security settings, access banned sites, install untrusted content, or perform any malware execution.

## 22.3 Potential User Problems

Due to the non-deterministic nature of dysarthric speech patterns, a user's exact speech type may not be fully captured by the model, leading to higher training overhead for personalization before the system becomes reliably usable for them. Users may experience misinterpretations requiring retries, which can cause frustration.

## 22.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

Variability in dysarthric speech may necessitate frequent retraining of ASR models.

Real-time performance varies between devices, depending on processing power and low-latency operation.

## 22.5 Follow-Up Problems

The local operation limits the ability to perform remote problem diagnosis if any issues arise.

# 23 Tasks

## 23.1 Project Planning

Breakdown of major tasks:

Our main tasks are centered on preparing the system by training and tuning the dysarthric ASR model, integrating the core command interpreter, designing the accessible interface, and conducting rigorous testing.

## 23.2 Planning of the Development Phases

A detailed schedule of development phases, milestones, and dependencies is outlined in the PoC and Development Plan.

# 24 Migration to the New Product

## 24.1 User Onboarding and Migration

The system shall support new users with no prior ASR and interpreter experience through an onboarding process.

The system shall support transitioning users migrating from an existing system by allowing for easy uploading of recordings of their speech training data.

### 24.2 Data Support Across Accounts

All previous transcripts, audio files, and personalization data shall be tied to secure user accounts.

Users shall be able to access their data from any supported device after logging in and authentication.

## 25 Costs

The primary costs for this product are related to machine learning computation and hosting. Development and training will initially leverage Compute Canada credits provided by the supervising research team, along with Colab Pro (CAD 13.99/month).

No external hosting costs are anticipated for the initial release; however, optional hosting may be required for commercial distribution and deployments. If hosting is included, approximate costs for cloud-based deployment on Google Cloud Platform (GCP) include: training and inference on GPUs ( $50 CAD/month for pilot use), storage ( $3 CAD/month), and minimal networking ( $2–5 CAD/month). Costs scale with the number of users and training frequency. All estimates are approximate and intended for planning purposes.

## 26 User Documentation and Training

### 26.1 User Documentation Requirements

The system shall provide clear, accessible documentation, including an instructional guide and FAQ. (As noted in Section 14b, supporting requirements already cover this in detail.)

### 26.2 Training Requirements

Training shall not require formal instruction; the documentation and interface shall support self-directed onboarding.

## 27 Waiting Room

The system shall display launch, loading, or processing pages during transitions to provide users with clear feedback and reduce confusion during waiting periods.

## 28 Ideas for Solution

### 28.1 Browser-Based Extensions

The product is primarily web-based. Future considerations could include desktop or mobile apps to broaden the scope and accessibility (see Section 13 for environment and interface requirements).

### 28.2 Personalized ASR Fine-Tuning

Consider incremental model tuning paired with real-time streaming of speech input. This could be an approach to adapt to individual speech patterns while minimizing the training time overhead, related to the adaptability requirements discussed in earlier sections.

### 28.3 LLM Command Mapping

User intent parsing via context-aware LLMs could improve natural language command interpretation, as mentioned in Section 13.

### 28.4 Noise Filtering

Beyond stationary noise, adaptive filtering techniques could be adapted to enhance recognition in busier environments. Section 13 already specifies stationary noise handling.

### 28.5 Open-Source Integrations

Potential use of browser automation or voice control frameworks can be investigated, as mentioned in Section 16 for off-the-shelf solutions.

### 28.6 Accessibility Enhancements

Additional visual, auditory, or haptic cues for feedback and help features could enhance usability. Section 14 highlights basic supportability requirements that these enhancements would build on.

### 28.7 Data Encryption

Anonymized logging and privacy approaches can balance model improvement with privacy in mind. Section 15 discusses privacy requirements in detail.

## 29 Requirements Phase-In Plan

The following plan outlines the order in which the system requirements (9.1–17.2) will be addressed, aligned with project phases. Prioritization ensures that core functional requirements and critical non-functional requirements are validated first, with refinements and extensions added in later phases. Timelines for each phase are detailed in the Development Plan.

### 29.1 Phase Definitions

- **Proof of Concept (PoC):** Validate feasibility of core functionality and critical performance metrics.
- **Revision 0:** Expand functionality, address usability, accessibility, and integrate initial security/privacy mechanisms.
- **Revision 1:** Final refinement, full productization, compliance, cultural, and maintainability requirements.

### 29.2 Requirements Phase-In

1. **9.1 Functional Requirements:** Core system behaviors including input processing and basic transcription.
   *Phase: PoC*
   *Rationale:* Core functionality must be demonstrated to validate feasibility. See Functional Requirements.

2. **10.1 Appearance Requirements:** Initial UI layout and interface elements.
   *Phase: Revision 0*
   *Rationale:* Visual interface can be refined after core functions are stable. See Appearance Requirements.

3. **10.2 Style Requirements:** Consistent fonts, colors, and spacing.
   *Phase: Revision 0*
   *Rationale:* Enhances readability and user experience once functionality is proven.See Style Requirements.

4. **11.1 Ease of Use Requirements:** Interface must be intuitive and straightforward.
   *Phase: Revision 0*
   *Rationale:* Important for usability testing, applied after functional PoC is complete. See Ease of Use Requirements.

5. **11.2 Personalization and Internationalization Requirements:** Language/voice adaptations.
   *Phase: Revision 1*
   *Rationale:* Applied later when system is functionally stable. See Personalization Requirements.

6. **11.3 Learning Requirements:** System should adapt to repeated user patterns.
   *Phase: Revision 1*
   *Rationale:* Machine learning or adaptation features require validated data pipelines. See Learning Requirements.

7. **11.4 Understandability and Politeness Requirements:** Feedback should be clear and respectful.
   *Phase: Revision 0*
   *Rationale:* Improves user experience during early testing. See Understandability and Politeness Requirements.

8. **11.5 Accessibility Requirements:** Interface supports assistive technologies.
   *Phase: Revision 0*
   *Rationale:* Early implementation ensures PoC testing is inclusive. See Accessibility Requirements.

9. **12.1 Speed and Latency Requirements:** System responds within acceptable time frames.
   *Phase: PoC*
   *Rationale:* Performance validation is crucial for feasibility. See Speed and Latency Requirements.

10. **12.2 Safety-Critical Requirements:** Ensure no harmful outputs or behaviors.
    *Phase: PoC*
    *Rationale:* Early validation prevents critical errors in PoC. See Safety-Critical Requirements.

11. **12.3 Precision or Accuracy Requirements:** Transcription must meet minimum WER/CER.
    *Phase: PoC*
    *Rationale:* Accuracy is the central PoC metric. See Precision or Accuracy Requirements.

12. **12.4 Robustness or Fault-Tolerance Requirements:** Handle errors gracefully.
    *Phase: Revision 0*

*Rationale:* Necessary for reliable system use beyond PoC. See Robustness or Fault-Tolerance Requirements.

13. **12.5 Capacity Requirements:** Support a small set of concurrent users/data.
    *Phase: Revision 0*
    *Rationale:* Ensure system can handle expected early usage. See Capacity Requirements.

14. **12.6 Scalability or Extensibility Requirements:** System should allow future growth.
    *Phase: Revision 1*
    *Rationale:* Added after functional validation for longer-term planning. See Scalability or Extensibility Requirements.

15. **12.7 Longevity Requirements:** Components maintainable over time.
    *Phase: Revision 1*
    *Rationale:* Stability and maintainability are late-stage concerns. See Longevity Requirements.

16. **13.1 Expected Physical Environment:** System works on typical office/lab computers.
    *Phase: PoC*
    *Rationale:* Early testing on target devices is essential. See Expected Physical Environment.

17. **13.2 Wider Environment Requirements:** Browser compatibility.
    *Phase: Revision 0*
    *Rationale:* Once core functionality works, broaden environment testing. See Wider Environment Requirements.

18. **13.3 Requirements for Interfacing with Adjacent Systems:** Integration points identified.
    *Phase: Revision 1*
    *Rationale:* External integration is late-phase after internal systems are stable. See Requirements for Interfacing with Adjacent Systems.

19. **13.4 Productization Requirements:** Ready for deployment.
    *Phase: Revision 1*
    *Rationale:* Combines all prior validations for release. See Productization Requirements.

20. **13.5 Release Requirements:** Packaging, distribution readiness.
    *Phase: Revision 1*
    *Rationale:* Ensures final delivery meets standards. See Release Requirements.

21. **14.1 Maintenance Requirements:** System maintainable.
    *Phase: Revision 1*
    *Rationale:* Implemented once core system is functional. See Maintenance Requirements.

22. **14.2 Supportability Requirements:** Technical support procedures.
    *Phase: Revision 1*
    *Rationale:* Late-stage requirement after main functionality. See Supportability Requirements.

23. **14.3 Adaptability Requirements:** System can be upgraded.
    *Phase: Revision 1*
    *Rationale:* Flexibility comes after base system is validated. See Adaptability Requirements.

24. **15.1 Access Requirements:** User roles and permissions.
    *Phase: Revision 0*
    *Rationale:* Early implementation secures initial PoC data. See Access Requirements.

25. **15.2 Integrity Requirements:** Prevent tampering or corruption.
    *Phase: Revision 0*
    *Rationale:* Ensures reliable data handling. See Integrity Requirements.

26. **15.3 Privacy Requirements:** Encrypt data in transit and at rest.
    *Phase: Revision 0*
    *Rationale:* Essential for user trust and legal compliance. See Privacy Requirements.

27. **15.4 Audit Requirements:** Track critical system events.
    *Phase: Revision 1*
    *Rationale:* Auditing implemented after basic functionality and privacy are validated. See Audit Requirements.

28. **15.5 Immunity Requirements:** Protection from external attacks.
    *Phase: Revision 1*
    *Rationale:* Security hardening is a late-stage activity. See Immunity Requirements.

29. **16.1 Cultural Requirements:** Respect cultural norms and sensitivities.
    *Phase: Revision 1*
    *Rationale:* Applied after all functional, UI, and content elements are stable. See Cultural Requirements.

30. **17.1 Legal Requirements:** Comply with laws and regulations.
    *Phase: Revision 1*
    *Rationale:* Legal compliance verified after security, privacy, and functionality are validated. See Legal Requirements.

31. **17.2 Standards Compliance Requirements:** Follow industry standards.
    *Phase: Revision 1*
    *Rationale:* Ensures credibility and certification readiness at final stage. See Standards Compliance Requirements.

### 29.3 Requirement Implementation Timeline

The phasing of requirements for VoiceBridge follows a structured, incremental schedule aligned with prototype maturity and evaluation cycles. The table below summarizes the planned order and target completion dates for each phase.
This phased plan ensures that feasibility and safety are validated early (PoC), usability and inclusivity are developed next (Revision 0), and full scalability, compliance, and maintainability are achieved in the final iteration (Revision 1).

## 30 Formal Specification

This section provides a simplified formal model of the VoiceBridge system using a state-based representation. The goal is to describe how the system transitions between major states during operation.

| Phase | Target Date | Focus and Included Requirements |
|---|---|---|
| **Proof of Concept (PoC)** | October–November 2025 | Implement core functionality: speech input capture, transcription (FR9.1, 12.1–12.3, 13.1), and safety validation (12.2). Establish baseline latency, accuracy, and system feasibility. |
| **Revision 0** | December 2025–January 2026 | Integrate user interface, accessibility, and initial security layers. Includes appearance (10.1–10.2), usability (11.1, 11.4–11.5), robustness (12.4–12.5), and privacy/integrity controls (15.1–15.3). Conduct early user testing. |
| **Revision 1** | February–April 2026 | Expand system capability and compliance. Add personalization (11.2–11.3), scalability (12.6–12.7), interoperability (13.3–13.5), supportability (14.1–14.3), advanced security (15.4–15.5), and cultural/legal alignment (16.1, 17.1–17.2). Prepare for deployment. |

## 30.1 System States

Let the system have the following states:

$$Q = \{Idle, Listening, Processing, Confirming, Executing, Error\}$$

The system starts in the *Idle* state and reacts to user or system inputs.

## 30.2 Inputs and Transitions

Inputs to the system are represented as:

$$\Sigma = \{Start, AudioInput, Confirm, Reject, Execute, Fail\}$$

The transition function $\delta(q, \sigma)$ describes how the system changes state:

$$\begin{aligned}
&\delta(Idle, Start) = Listening \\
&\delta(Listening, AudioInput) = Processing \\
&\delta(Processing, Confirm) = Executing \\
&\delta(Processing, Reject) = Listening \\
&\delta(Executing, Execute) = Idle \\
&\delta(Executing, Fail) = Error
\end{aligned}$$

## 30.3 Output Behavior

Each state produces a specific output:

$$\begin{aligned}
&Idle \rightarrow \text{System ready prompt} \\
&Listening \rightarrow \text{Microphone active} \\
&Processing \rightarrow \text{Speech to text conversion} \\
&Confirming \rightarrow \text{User verification message} \\
&Executing \rightarrow \text{Command execution feedback} \\
&Error \rightarrow \text{Error or retry prompt}
\end{aligned}$$

### 30.4 Mapping to Requirements

This formal model represents the main workflow of VoiceBridge and aligns with the following functional requirements:

- FR1: Accept Speech Input ($Idle \rightarrow Listening$)
- FR2: Convert Speech to Text ($Listening \rightarrow Processing$)
- FR3: Confirm Transcription ($Processing \rightarrow Confirming$)
- FR4: Map Intent to Command ($Confirming \rightarrow Executing$)
- FR5: Execute Command ($Executing \rightarrow Idle$)

This simplified formalization shows how user input moves through each stage of the system in a predictable, testable way, ensuring clarity and correctness of behavior.

## Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. **What went well while writing this deliverable?**
   Organizing the SRS using the Volere template went surprisingly smoothly. Breaking down each requirement and writing clear rationales helped us think through the project in a structured way. Separating likely-to-change requirements from unlikely ones also made the document more readable and easier to navigate.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**
   Deciding which requirements might evolve over time versus those that would remain stable was a bit challenging. We resolved this by discussing potential future scenarios for the product and weighing Dr. Brodbeck's expert guidance, which made our decisions much more confident.

3. **How many of your requirements were inspired by speaking to your client(s) or their proxies?**
   Our requirements mostly came from our supervisor, Dr. Brodbeck, who is a linguistics expert and acts as our secondary stakeholder. We didn't consult end-users directly, so the requirements reflect expert advice on accessibility, usability, and system functionality.

4. **Which of the courses you have taken, or are currently taking, will help your team succeed with this project?**

   - **SFWRENG 3RA3 – Software Requirements and Security:** Helped with writing clear Functional and Non-Functional Requirements and thinking about security, privacy, and reliability.
   - **SFWRENG 4HC3 – Human-Computer Interfaces:** Provided a foundation for usability, stakeholder engagement, and accessibility considerations.
   - Other courses in AI and Machine Learning are also valuable for speech recognition and natural language understanding.

5. **What knowledge and skills will the team need to successfully complete this project?**
   We need to improve on:

   - Speech and linguistics for accessibility.
   - Web development, particularly browser extensions.
   - Machine learning for speech recognition.
   - Accessibility standards and usability testing.

- Clear documentation and team coordination.

6. **How will each team member acquire these skills or knowledge?**

   - **Speech and linguistics:** Weekly guidance from Dr. Brodbeck and reading related literature.
   - **Web development/browser extensions:** Hands-on prototyping, online tutorials, and sharing coding tasks among team members.
   - **Machine learning:** Experimenting with pre-trained speech models and following online courses.
   - **Accessibility standards:** Reviewing WCAG guidelines and testing our prototype with assistive technologies.